- *Powder conditioning* is an important processing step in P/M and involves mechanical, thermal or chemical treatments or alloying of the powders.
- The significance of powder treatment can be understood from the fact that the powder size, shape and surface conditions have a significant **influence on the subsequent processing.**
- In powder conditioning, the powders prepared by various methods are subjected to a variety of treatments to improve or modify their physical or chemical characteristics.
- For instance, **wet powders are dried** to avoid any problems during subsequent processing, especially sintering.
- Physical characteristics like **particle size, size distribution and shape may be varied by milling**.

POWDER TREATMENTS

- Powders are broadly classified into two major groups:
- **elemental and prealloyed powders.**
- Elemental powders refer to powders of a single metallic element.
- Prealloyed powders consist of more than one element and are made by alloying elemental powders during the manufacturing process itself.
- In prealloyed powders, all the particles have the same nominal composition and each particle may be considered as a miniature ingot.
- Depending upon the nature of the powder, whether elemental or prealloyed, as well as on the type of application, powders are subjected to a variety of treatments.

(i) Drying to remove moisture.
(ii) Grinding/crushing to obtain fine sizes (e.g. powder agglomerates).
(iii) Particle size classification to obtain the desired particle size distribution.
(iv) Annealing to improve compaction.
i. Mixing of different powders for premixes, and blending of powders and powder mixes to obtain a homogenous mix.
ii. Lubricant addition for compaction of powders.
iii. Powder coating.

Cleaning of Powders

- Cleaning of powders refers to the removal of contaminants, solid or gaseous, from the powders.
- This is particularly important in the case of high-strength parts requiring good fatigue or creep properties as in turbine parts.
- Solid or particulate contaminants come from several sources like nozzles (in atomization) or crucible linings.
- They interfere with compaction and sintering, preventing proper mechanical bonding
- **Particle separators as well as electrostatic separation techniques are used to remove** non-metallic particles from superalloy powders.
- Gaseous impurities like hydrogen and oxygen get into powders during processing, storage or handling if proper care is not taken.

- **Finer the powders, more will be the level of contamination**, owing to the very large surface area of the powders.
- These impurities can form undesirable oxides during processing at high temperatures or may remain trapped within the material as porosity, reducing the strength and dynamic properties of the sintered part.
- **Degassing techniques like** cold and hot static degassing, cold and hot dynamic degassing are used to remove adsorbed gases from the powders.
- **Lubricants,** added to powders to facilitate compaction, **if not removed properly prior to sintering can cause problems during sintering**.
- In general, the type and amount of impurity will depend on the powder manufacturing process used as well as the handling methods.

Grinding

- The primary aim of grinding is comminution of powders.
- Grinding is accomplished using variety of equipment such as ball mill or attritor.
- Main aim is to get fine powder size

Powder Classification and Screening

- P/M part production requires specified particle size distributions in order to ensure uniform properties in the sintered part.
- For example, in the case of **porous filters**, it is essential to control the **particles size within a very narrow size range**.
- Generally, powder production processes yield a wide band of particle size distributions.
- The influence of several process variables can cause the size distributions and average particle sizes of the resulting powders to show a wide variation from one lot to another.
- These variations can lead to variation in properties of the powder and sintered part. Classifying and screening methods are employed to obtain the desired particle size distributions with particle sizes within specific limits.
- **Necessary to control the particle size distribution within specified limit.**

Blending and Mixing

- Blending and mixing of powders **ensure a uniform distribution of different constituents like powders**, additives and lubricants, and facilitate good compaction and sintering.
- These operations are necessary, especially if the application involves a multi-component system.
- **Segregation of different components/fine and coarse particles occur during transportation and/or filling** operations.
- Segregation occurs in both multi-component and single-component systems due to differences in particle size, shape, density and other powder characteristics.
- Blending and mixing are often carried out under controlled conditions (inert gas atmosphere or using a liquid medium) to avoid contamination of the powder.

Blending

- Blending is the process in which powders of the **same nominal composition but having different particle sizes** are intermingled.
- Blending of metal powder is an important powder treatment and is done

  (i) to obtain a uniform distribution of particle sizes, i.e. powders consisting of different particle sizes are often blended to minimize porosity, and

  (ii) for intermingling of lubricant with powders to modify metal-powder interaction during compaction and facilitate the compaction.

Mixing

- Mixing refers to the process of **combining powders of different chemistries** such as elemental powder mixes (copper-tin) or metal-non-metal powders (e.g. friction materials).
- P/M enables the mixing of various materials and making them into finished **parts that would be otherwise difficult or impossible to make.**
- In mixing, additives like graphite and lubricants are also used to obtain a homogeneous mixture.
- A mixing operation is a must before compaction of powders.
- In case of mixing of two or more metal powders, **particle characteristics** like size and size distributions play a role in **influencing the degree of mixing**.
- This is important because many P/M applications use **metal and non-metallic constituents of different specific gravities, particle size and shapes to give optimum properties.**

- Mixing may be done either in **dry or wet condition.**
- Wet mixing is used to produce a more fine and uniform mixture of powder particles.
- Liquid mediums like alcohol, acetone, benzene or distilled water are used as milling medium in wet milling.
- Various types of blenders and mills are employed for mixing.
- Ball mills or rod mills are employed extensively for mixing hard metals such as carbides.
- **Spray drying is also used for mixing of constituents in the case of oxide dispersion strengthened alloys and cemented carbides.**

- The use of powder mixtures rather than **prealloyed powders** is attributed to both economic and technical factors.

- **Powder mixtures are often less expensive and have better compacting properties.**

- Soft and ductile materials may not be amenable to milling because of possible work hardening.

- **The mixed or blended powders are subsequently sieved to remove any foreign particles, powder aggregates and lubricant.**

Factors affecting mixing

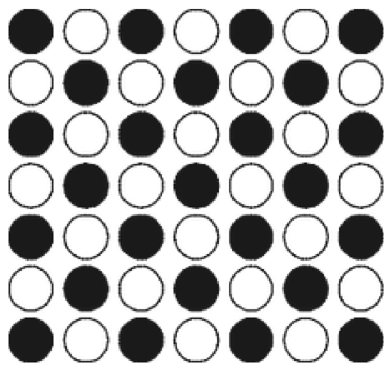
- Type of mixers (V-type, double cone or cylindrical).
- Volume of the mixers.
- Construction material and surface finish.
- Volume ratio of components.
- Volume ratio of mixers.
- Type of powders and their characteristics.
- Time of mixing.
- Temperature of mixing.
- Type of mixing (wet or dry mixing).

Particle characteristics affecting the mixing characteristics of powders

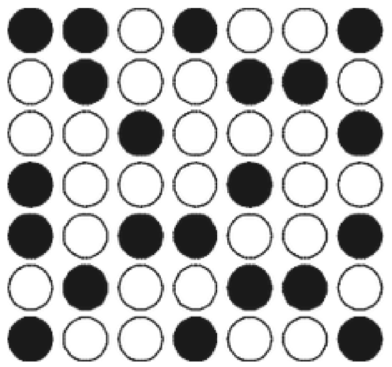
(i) The specific gravity of the component powder.
i. Particle sizes of individual components.
ii. Particle shape and structure.
iii. Particle surface conditions.

Mixing must be controlled to produce a uniform distribution of ingredients having specific particle characteristics.
The properties of the powder mixture such as apparent density and flow will depend on the mixing time and intensity of mixing of the powder and lubricant.
Mixing time must be optimized experimentally to achieve the desired degree of mixing.

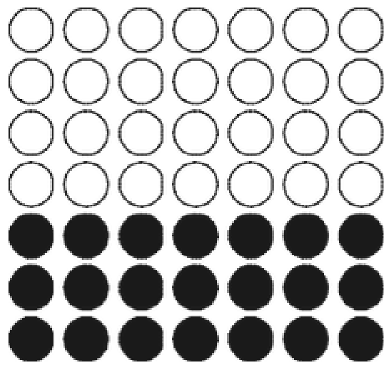
Perfectly mixed



Randomly mixed



Unmixed (no mixing)



Mechanisms of mixing

(i) Convective mixing: transfer of one group of particles from one location to another.
(ii) Diffusive mixing: movement of particles on to newly formed surface.
(iii) Shear mixing: deformation and formation of planes within the powders.

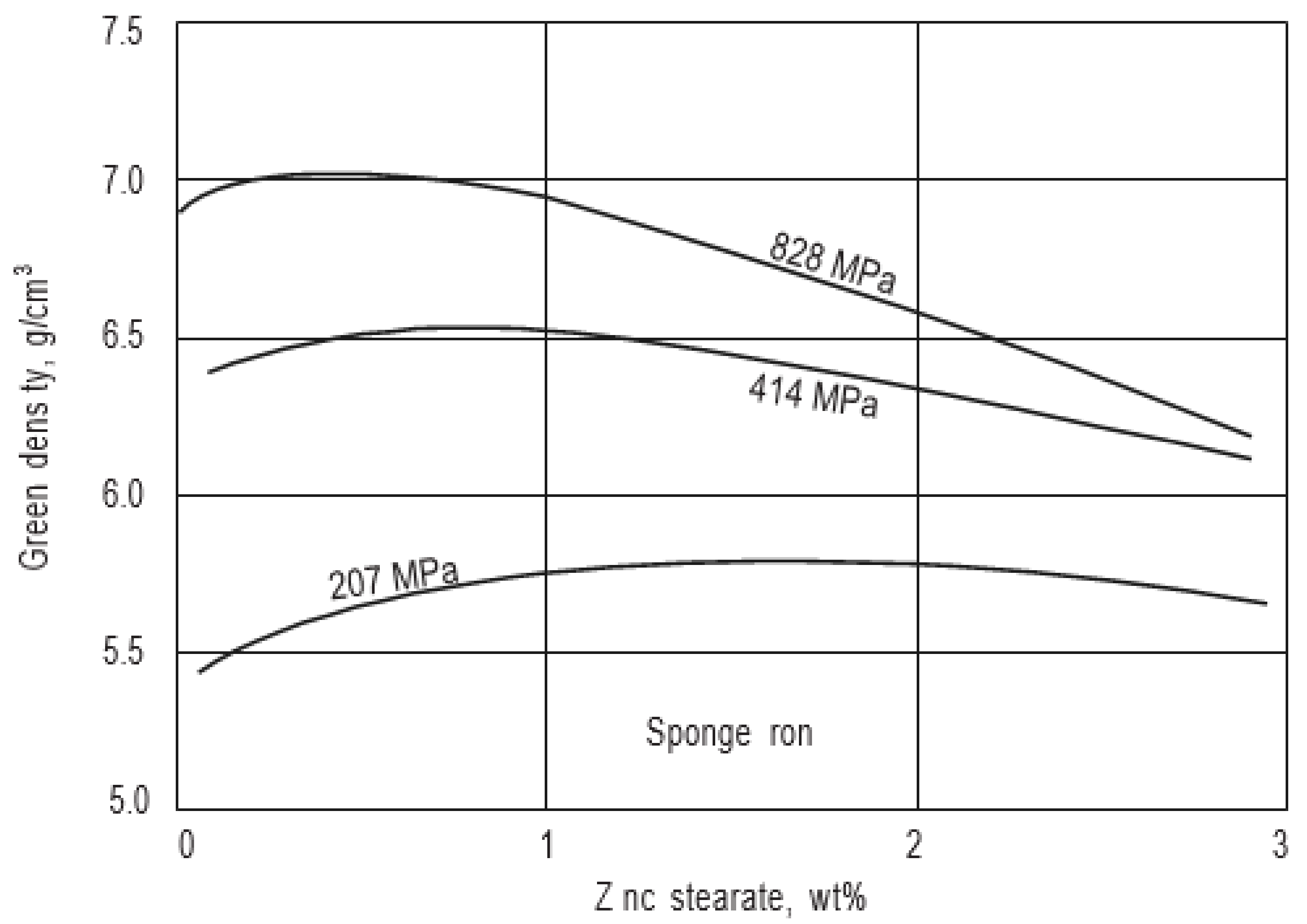
Lubrication

- Amount of lubricant added in blending or mixing will not exceed more than 2% by weight of the charge.
- For large and complex shapes may necessitate the use of larger amount of lubricant to be incorporated to facilitate easy removal of the green compact.
- Usually lubricants are introduced as fine powder and mixed with the metal or alloy powders.
- **The main function of the lubricant is to minimise the inter-particle friction as well as friction between powders and the die walls/core rods.**
- This helps to obtain a uniform and desired density in the compact.
- Lubricants are chosen such that they **attach themselves strongly to the metal surfaces and are not easily penetrated**.

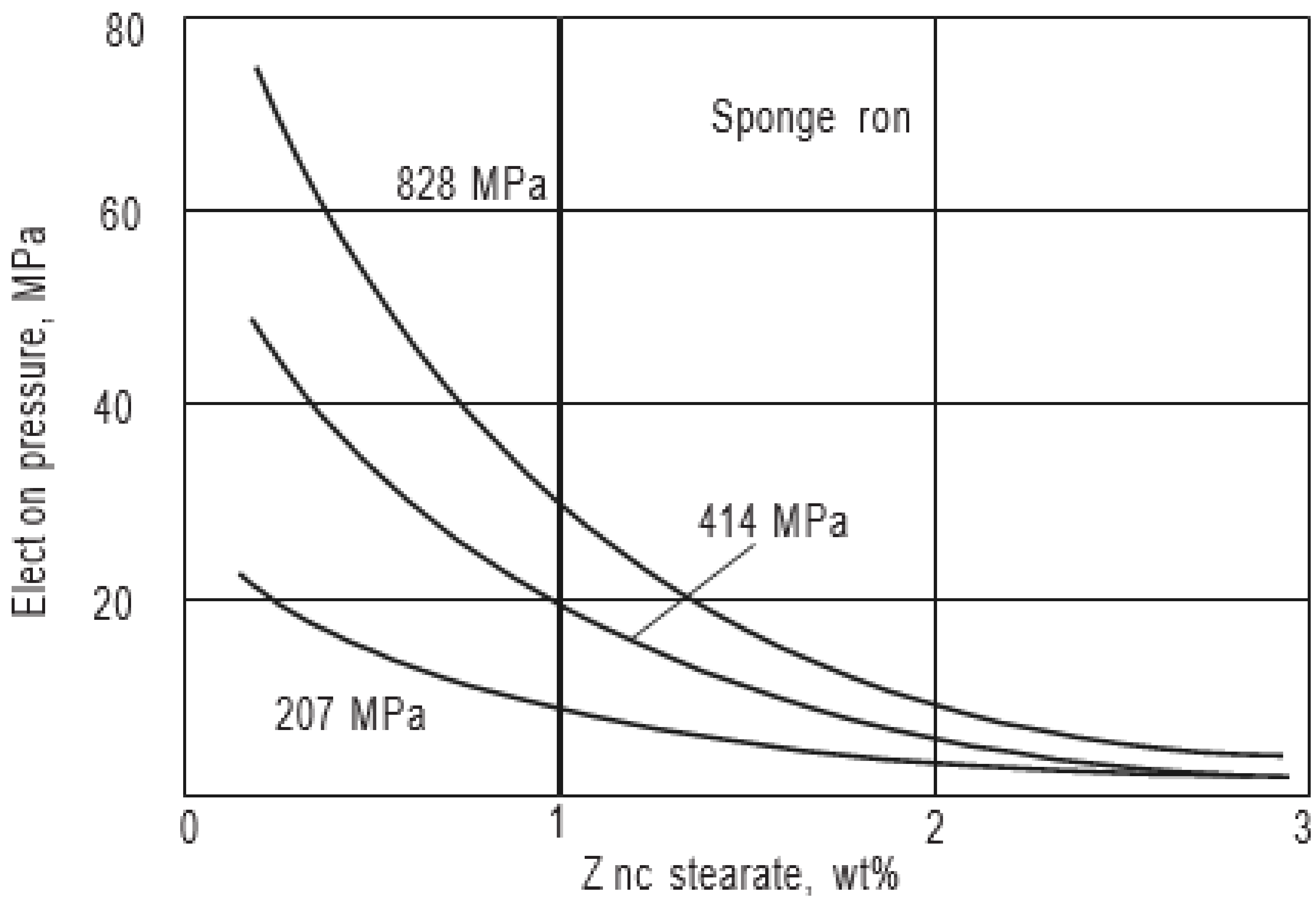
- Lubricants (as well as additives like graphite) also help to achieve the required green strength and controlled porosity in the finished part.
- Without proper lubrication, the pressure necessary to eject the compact from the die would be very high and may lead to cracking of the compact.
- In addition, use of lubricant also prevents die wear and contributes to the saving in operational costs.
- However, the type and amount of lubricant must be chosen carefully to avoid problems during subsequent processing: for example, care should be taken in the selection of lubricants since the **strength of green as well as that of the sintered part will be affected adversely** if any residue is left in the compact after their removal.
- Lubricants can also cause problems like **blistering and carburization of the** sintered part.
- Commonly used lubricants are **zinc stearate, stearic acid, lithium stearate, and synthetic waxes**.

Effect of powder lubrication on the green properties of pressed sponge iron.

- The green density is shown as a function of the amount of zinc stearate for three compaction pressures



Influence of the amount of lubricant (zinc stearate) on the force required for ejection of a compact.



Heat Treatment of Powders

- A preliminary heat treatment is usually carried out usually before mixing or blending the metal powders.
- Some of the important objectives of heat treatment are:

(i) *Improving the purity of the powder*

(ii) *Improving the softness of powders*

(iii) *Modification of powder characteristics*

Improving the purity of the powder

- **Reduction of surface oxides** from the powders by annealing in hydrogen or other reducing atmosphere.
- Powders produced by methods like atomization, reduction and electrolysis contain dissolved gases such as hydrogen and oxygen as impurities.
- **Annealing of powders** is done before pressing to **remove these dissolved gases** as well as other impurities like carbon.
- Heat treatment is usually done under a protective atmosphere like hydrogen or under vacuum.

Improving the softness of powders

- Powder agglomerates are generally crushed to obtain fine powders, while many powders are made by **milling, crushing or grinding of bulk materials.**

- The powders invariably get **work hardened** during processing.
- The powders are **annealed usually in a reducing atmosphere** such as hydrogen to eliminate the effects of work hardening.
- Annealing is done in a regular sintering type furnace and the **temperature usually is kept low to avoid fusion of the particles.**

Modification of powder characteristics

- **Apparent density** of the powders can be modified to a higher or lower value by altering the temperature of treatment.

- This treatment results in a change in the pressure needed for compaction.

Pyrophoricity

- *Pyrophoricity* is the susceptibility of metal **powders to rapid oxidation.**
- This property depends upon the chemical and physical characteristics of powders such as composition, size and surface area.
- Powders of **aluminium, iron and magnesium are highly pyrophoric** in nature whereas powders of copper and tungsten are not pyrophoric even at very fine sizes.
- **Uranium** can be pyrophoric at much larger sizes than iron and copper.
- Alkali metals like **lithium**, for instance, can react with water to liberate hydrogen and the liberated heat is sufficient to cause **explosion**.
- Lithium can also react with and **burn in nitrogen.**
- **Iron and copper powders show moderate explosivity**, while cobalt, lead and molybdenum show low explosivity.
- In dealing with powders that are pyrophoric in nature, it is necessary to follow certain **safeguards against possible explosion.**

Important requirements for safe powder handling

Providing **appropriate extinguishing agents** in locations where pyrophoric/ combustible materials are stored, handled or processed. **Dry sand, dry sodium chloride or soda ashes** are the recommended extinguishing agents in case of fire hazards. Some pyrophoric metals react with water or foams and cause explosion. These may require special powders/use of inert gases. In **large storage tanks for powders**, any **unoccupied volume must be filled using nitrogen or inert gases like argon or helium** to avoid reaction with air.

Use of **moisture-free processing atmosphere**, use of a fireproof enclosure, preferably free from oxygen.

Processing should be preferably done under **inert gas atmosphere.**

Use of **non-sparking tools during** handling operations.

Keeping easily combustible materials (wood, paper) away from the powder storage/handling area.

Contd….

- Avoiding use of open flame in the powder storage and handling areas.
- Avoiding dust build-up and storing flammable gases away from the storage area.

Toxicity

- Toxic effects can be caused due to skin contact, inhalation or ingestion.
- Powders like lead, nickel and beryllia are highly toxic while others like aluminum and iron show very little toxicity.
- The effects may be varying from irritation of eye and skin to nausea, vomiting, respiratory problems and even blood poisoning (e.g. nickel).

Precautions while powder handling

(i) Use of protective gear, gloves, respiratory masks, protective clothing and shoes.
(ii) Use of well-ventilated storage/workplace.
(iii) Careful handling/disposal of fines, effluents or waste fluids.